

Assignment 1: Scanning

COMP 4060 – Compiler Construction I
September 2026

[**Note:** *This assignment is adapted from the original version authored by Prof. Douglas Thain.*]

Problem description

Note: *In the following, “you” refers generically your group.*

In this assignment, you will use the Flex scanner generator to construct a scanner for the B-Minor 2025 programming language which reads in a source file and produces a listing of each token one by one, annotated with the token kind (identifier, integer, string, etc.) and the location in the source.

If invalid input is discovered, produce an error message, recover from the error, and continue.

You will find a folder containing multiple test input files (in B-Minor) in the GitHub repository referenced below. You should create any additional test input files that you consider necessary to exercise tricky corner cases of comments, strings, escape characters, and so forth, that might not be covered by the supplied test cases.

Division of labor and submission procedure

You will work in teams as defined by your Self-Enrollment Group. If you split up the deliverables, consider dividing the implementation portion and the testing portion.

To turn in your code and test cases, use the following procedure, which will be the same for all assignments this semester:

1. Your code and test cases (including any needed files, such as Makefiles) should be in a directory (folder) called “<GroupName>_CC_A1” (for example, “Group1_CC_A1”).
2. In the parent directory of your A1 directory, create a `.tar.gz` file (commonly called a “[tarball](#)”), or a `.zip` file, that contains your A1 directory (e.g., “Group1_CC_A1.zip”).
3. In the page for Assignment 1 on Canvas, use the Submit button to upload your `.tar.gz` or `.zip` file.

Note that only one person in the team needs to submit the code; when you submit, give the name of your partners in the submission comments field on Canvas.

Resources

- [B-Minor 2025 Language Overview](#)
This specification differs slightly from the one given in the required textbook (Thain, Appendix B). See the language overview for a list of differences.
- [Flex – a scanner generator](#)
Manual for the GNU Flex automatic scanner generator tool, which is faster than the older `lex` tool that is a standard UNIX system program.
More recent manual, for current version on CS servers (flex 2.6.4):
 - https://www.ida.liu.se/~TDDE66/laboratories/instructions/_static/flex/index.html
- <https://github.com/dthain/compilerbook-starter-code>
Contains starter code for the B-Minor compiler project.
Students in COMP.4060 are **required** to use these header files as the basis for your work. Trust us, it will get you started on the right track, and make it much easier for the instructor and TA to help you with any difficulties.
(**Note:** Not all of the header files provided in the starter code will be used in each stage of your compiler.)
- <https://github.com/dthain/compilerbook-examples>
Contains the major code examples from the textbook, as well as a folder containing sample test files for each phase of the compiler.
(**Note:** The “starter-code” folder contained in this examples folder is an older version than the starter-code folder linked above.)